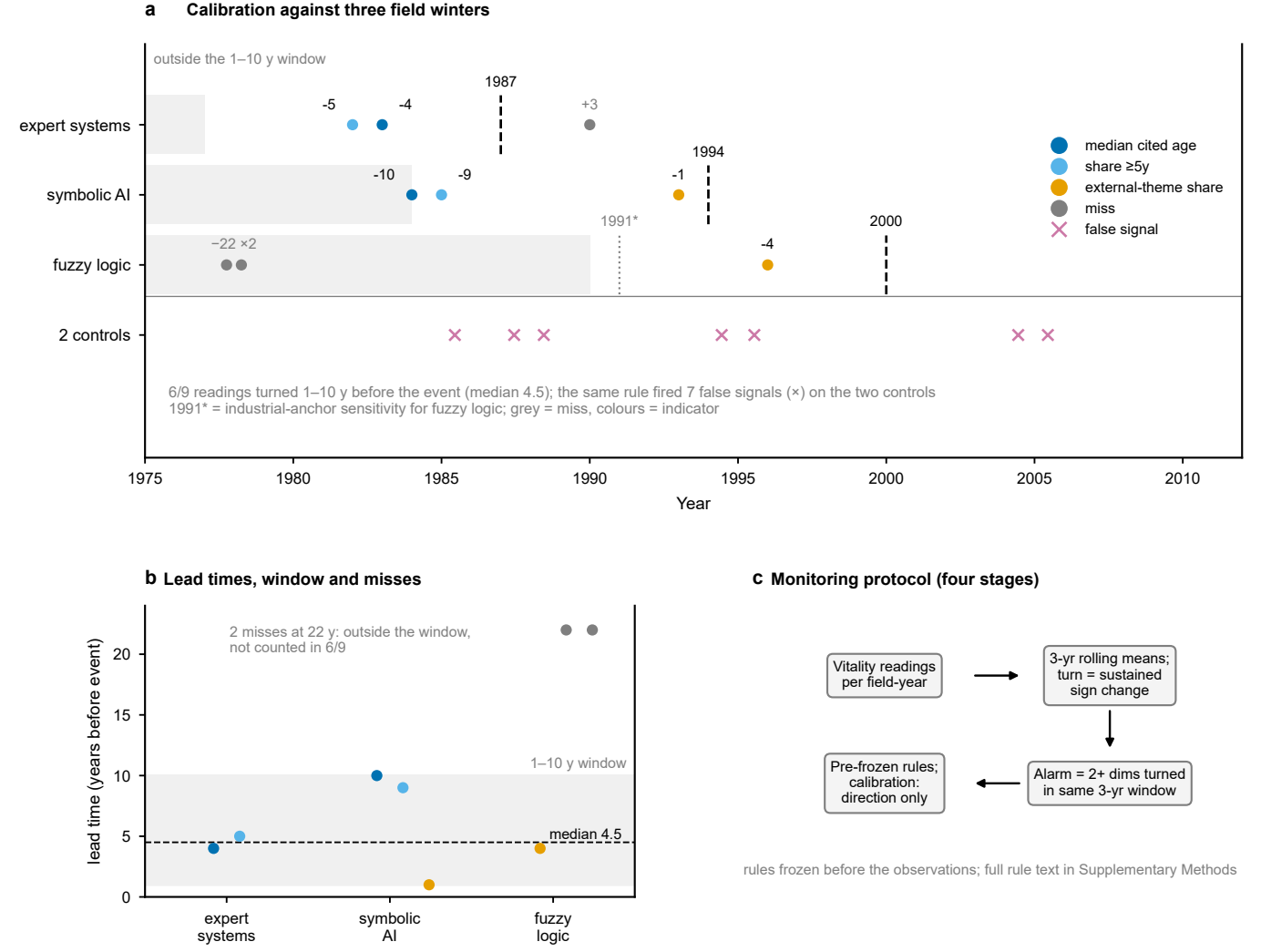

a Calibration against three field winters
outside the 1–10 y window
1987
1994
1991*
2000
-5
-4
+3
-10
-9
-1
−22 ×2
-4
expert systems
symbolic AI
fuzzy logic
2 controls
median cited age
share ≥5y
external-theme share
miss
false signal
6/9 readings turned 1–10 y before the event (median 4.5); the same rule fired 7 false signals (×) on the two controls
1991* = industrial-anchor sensitivity for fuzzy logic; grey = miss, colours = indicator
1975
1980
1985
1990
1995
2000
2005
2010
Year
b Lead times, window and misses
2 misses at 22 y: outside the window, not counted in 6/9
1–10 y window
median 4.5
lead time (years before event)
0
5
10
15
20
expert systems
symbolic AI
fuzzy logic
c Monitoring protocol (four stages)
Vitality readings per field-year
3-yr rolling means; turn = sustained sign change
Alarm = 2+ dims turned in same 3-yr window
Pre-frozen rules; calibration: direction only
rules frozen before the observations; full rule text in Supplementary Methods